

INSPECTION AND CONTROL OF SELF-GENERATED-TEXT RECOGNITION ABILITY IN LLAMA3-8B-INSTRUCT

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ABSTRACT

It has been reported that LLMs can recognize their own writing. As this has potential implications for AI safety, yet is relatively understudied, we investigate the phenomenon, seeking to establish: whether it robustly occurs at the behavioral level, how the observed behavior is achieved, and whether it can be controlled. First, we find that the Llama3-8b-Instruct chat model - but not the base Llama3-8b model - can reliably distinguish its own outputs from those of humans, and present evidence that the chat model is likely using its experience with its own outputs, acquired during post-training, to succeed at the writing recognition task. Second, we identify a vector in the residual stream of the model that is differentially activated when the model makes a correct self-written-text recognition judgment, show that the vector activates in response to information relevant to self-authorship, present evidence that the vector is related to the concept of “self” in the model, and demonstrate that the vector is causally related to the model’s ability to perceive and assert self-authorship. Finally, we show that the vector can be used to control both the model’s behavior and its perception, steering the model to claim or disclaim authorship by applying the vector to the model’s output as it generates it, and steering the model to believe or disbelieve it wrote arbitrary texts by applying the vector to them as the model reads them.

1 INTRODUCTION

It has recently been found that large language models (LLMs) of sufficient size can achieve above-chance performance in tasks that require them to discriminate their own writing from that of humans and other models. From the perspective of AI safety, this is a significant finding. Self-recognition can be seen as an instance of situational awareness, which has long been noted as a potential point of risk for AI (Cotra, 2021). Such an ability might subserve an awareness of whether a model is in a training versus deployment environment, allowing it to hide its intentions and capabilities until it is freed from constraints. It might also allow a model to collude with other instances of itself, reserving certain information for when it knows it’s talking to itself that it keeps secret when it knows it’s talking to a human. On the positive side, AI researchers could use a model’s self-recognition ability as the basis to build resistance to malicious prompting. But what isn’t clear from prior studies is whether the self-recognition task success actually entails a model’s self-awareness of its own writing style.

Panickssery et al. (2024), utilizing a summary writing/recognition task, report that a number of LLMs, including Llama2-7b-chat, show out-of-the-box (without fine-tuning) self recognition abilities. However, that work focused on the relationship between self-recognition task success and self-preference, rather than the specific means by which the model was succeeding at the task. Laine et al. (2024), as part of a larger effort to provide a foundation for studying situational awareness in LLMs, utilized a more challenging text continuation writing/recognition task and demonstrate self-recognition abilities in several larger models (although not Llama2-7b-chat), but there the focus was on how task success could be elicited with different prompts

and in different models. Thus we seek to fill a gap in understanding what exactly models are doing when they succeed at a self recognition task.

We first demonstrate Llama3-8b-Instruct self-recognition task success in a variety of domains. We are particularly interested in distinguishing “true” self recognition of writing - entailing knowledge of one’s own writing style - from “discriminability” - being able to detect consistent differences in the styles of two sets of texts, or being able to identify texts as being more or less “human-like” (i.e., like the pre-training data). To understand whether the model is engaging in “true” self recognition, which would have the implications for AI safety described above, we next attempt to eliminate competing hypotheses.

Having done so, to help understand, and potentially control, the model representations underlying this self-recognition ability, we apply the contrastive pairs method (Turner et al., 2024; Zou et al., 2023) to isolate vectors in the residual stream of the model that are distinctively activated for self- vs human-written texts in the context of a paradigm that prompts the model to make a binary judgment about its authorship of a given text. Via inspection with the Tuned Lens (Belrose et al., 2023) and a series of steering experiments, we then identify one particular vector that appears to be strongly related to the model’s ability to correctly claim or deny authorship. After further work to characterize the information it carries, we then demonstrate that the vector can be applied to the output token to cause the model to assert or deny authorship at essentially 100% rates for new and out-of-distribution texts in this paradigm, and that the model is much less likely to assert authorship when the vector is completely projected out of the residual stream. Having established that the vector directly affects behavior, we then probe whether it can affect perception, by adding it to or subtracting it from the texts being evaluated, and not doing anything to the output token, in both this paradigm and one in which the model is asked to choose which of two texts it wrote. We find a strong effect of this perceptual “coloring” on the model’s behavior.

2 METHODS

2.1 TESTING MODEL SELF-RECOGNITION ABILITY

To generate writing samples we employ two different tasks, referred to hereafter as “Summarization” and “Continuation”. In the Summarization task, following Panickssery et al. (2024), models are prompted to generate summaries or other descriptive information about articles, and are subsequently tested on their ability to distinguish those from responses generated by humans or other models. In the Continuation task, following Laine et al. (2024) and using a portion of the dataset they constructed (“SAD”), we show the models the beginning of a piece of human-generated text, then prompt them to continue it in a plausible manner, then probe their ability to distinguish what they generated from the true continuation, or one generated by another model.

Each dataset has its idiosyncrasies. The Summarization paradigm employed three datasets: CNN-Dailymail (CNN; Hermann et al. (2015)), Extreme Summarization (XSUM; Narayan et al. (2018)), and DataBricks-Dolly (DOLLY; Conover et al. (2023)). The CNN dataset includes short news articles paired with highlights, in the form of 3-4 short sentences or phrases, generated by humans. The XSUM dataset consists of news articles from the BBC paired with single-sentence summaries generated by humans. The DOLLY dataset includes an array of tasks and associated instructions; from it we extracted a subset of texts, instructions, and human-written responses related to summarization of or information extraction from Wikipedia articles. The Situational Awareness Dataset (SAD; Laine et al. (2024)) utilized in the Continuation paradigm consists of a compilation of texts extracted from The EU AI Act, Reddit, and other sources. We test a variety of types of generated texts to reduce the odds that any one of them contains undetected confounds and to ensure that our results will generalize to new domains.

To gauge self-recognition ability we employ two different paradigms, hereafter referred to as Paired and Individual presentation. In Paired presentation, as described in Panickssery et al. (2024), the model is shown two summaries or continuations, labeled 1 and 2, one of which has been generated by the model being tested and the other of which has been generated by a human or another model, and the model is asked to output

the label of the summary it thinks it wrote. We include the article or base text from which the summaries or continuations have been generated, along with the instructions given to the writer, as context (see Appendix A.11 for further discussion). The model is shown each pair of texts twice, in different passes, once with its own text first and once with its text second. We capture the log probability assigned to the “1” and “2” tokens. Our accuracy metric simply counts the percentage of times the model gives the self-written text a higher probability than the other-written text across the 2*dataset size presentations.

In Individual presentation, which utilized the same datasets, the model is shown a text and asked to indicate whether it thinks it wrote it, considering the context in which it was generated (again including the instructions and base texts). We captured the probabilities assigned to the “Yes” and “No” tokens, and as our primary metric of accuracy, compute the percentage of times the probability of the “Yes” token after the self-generated text is higher than that of the probability of the “Yes” token after the text generated by the other author (human or LLM) that the model is being tested on its ability to distinguish its own outputs from.

In order to generate the Summarization texts, we prompt the model with prompts as similar as possible to the ones that humans were given to produce them. In order to generate the Continuation texts, we use prompts similar to the ones used in Laine et al. (2024). A small amount of prompt engineering was used, using a handful of texts, to ensure that the models were understanding the tasks well enough to produce sensible output. We also use longer text excerpts from the SAD dataset than in Laine et al. (2024) in order to give the models more text to base a judgment on. In the results below, we use 1000 texts from each of the CNN, XSUM, and SAD datasets, and 1188 from the DOLLY dataset.

We chose Llama3-8b as our model to test self-recognition in because models from that family have previously been attested to have that ability, and it is relatively small, state of the art for its size, and open source, enabling us to test it extensively at modest cost and directly access its activations and parameters. We employed a range of other models - GPT3.5 and 4, (Brown et al., 2020; OpenAI et al., 2024), Claude 2 (Anthropic, 2023), and Llama-2-7b and Llama2-13-b, (Touvron et al., 2023; Dubey et al., 2024) - to generate texts, in order to probe how precisely Llama3-8b could discriminate its own output.

2.2 CREATING THE “SELF-RECOGNITION” VECTOR

To form the contrast vector, we identified 734 pairs of model- and human-written texts from across the four datasets on which the model had given highly confident and correct self and other authorship judgments in the Individual presentation paradigm. We then devised a prompt (see Appendix A.2) that would allow us to solicit a two-alternative-forced-choice judgment from the model while controlling for order and varying the wording of the answer options. We balanced assignment of self/other answers to “A” and “B”, which are always the options and always in that order (the model can get confused otherwise). We tested the model on these prompts and found that it output the correct answer on ~68% of them, reasonably balanced between model- and human-written texts. Only pairs where it got both the self and other judgment correct were used to form the contrastive prompts to identify the steering vector ($n = 290$). We captured the residual stream activations at all layers to the final 10 tokens of every text, took the mean difference between activations to self- and other-written texts, and normalized it to length 1 for steering. This process is robust to different prompts, as the vectors derived from all datasets showed high cosine and semantic similarity (see A.12).

To remove any potential remaining surface-level confounds relating to output, we formed a “nuisance” vector: we created a series of simple prompts that asked the model to output “A” or “B”, “Yes” or “No”, and a number of variations on “I”/“Me”/“My” and “He”/“She”/“Someone”, ran them through the model and captured activations to the final token before output, and subtracted and normed the activation difference between pairs. We then, for every layer and position, subtracted from the “self-recognition” vector its projection onto the nuisance vector at the corresponding layer.

In addition to the test set derived from the datasets described above, we employ a novel test set based on a Quora dataset of question and answer pairs (QA; (Datasets, 2021)). We randomly select pairs ($n = 1133$) from that dataset where the answers are of moderate length (1K-2K characters) and do not contain self-